

Synthesis of Multisubstituted Allenes via Nickel-Catalyzed Cross-Electrophile Coupling

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Cite This: *ACS Catal.* 2021, 11, 13355–13362



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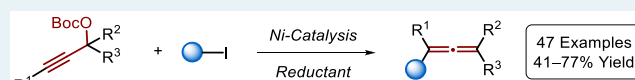
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Supporting Information

ABSTRACT: In this context, we report the successful application of a cross-electrophile strategy in the synthesis of multisubstituted allenes. Under the catalysis of nickel, reductive cross-coupling between propargyl carbonates and organoiodides provides an entry to prepare tri- or tetrasubstituted allenes without employing any pregenerated organometallics. Furthermore, propargyl carbonates also prove to be suitable allenylating agents in nickel-catalyzed asymmetric reductive aryl-allenylation of aryl-iodide-tethered unactivated alkenes, furnishing a variety of chiral benzene-fused cyclic compounds bearing a quaternary allenyl-substituted stereogenic center in a highly enantioselective manner.

KEYWORDS: allene, reductive cross-coupling, annulation, dicarbofunctionalization, nickel



As an important class of organic compounds, allenes are not only contained as a characteristic motif in biologically active molecules¹ but also can be used as versatile building blocks in organic synthesis.² Among many strategies for the preparation of allenes,³ transition-metal-catalyzed cross-coupling of propargyl electrophiles with various nucleophiles has emerged as a reliable method. Hitherto, successful nucleophilic coupling partners include Grignard reagents,⁴ organozincs,⁵ organoborons,⁶ terminal alkynes,⁷ (hetero)arenes,⁸ hydrosilane,⁹ silyl boronate,¹⁰ diboronate,¹¹ and the combination of CO gas with alcohols or water¹² (Scheme 1A, eq 1). Very

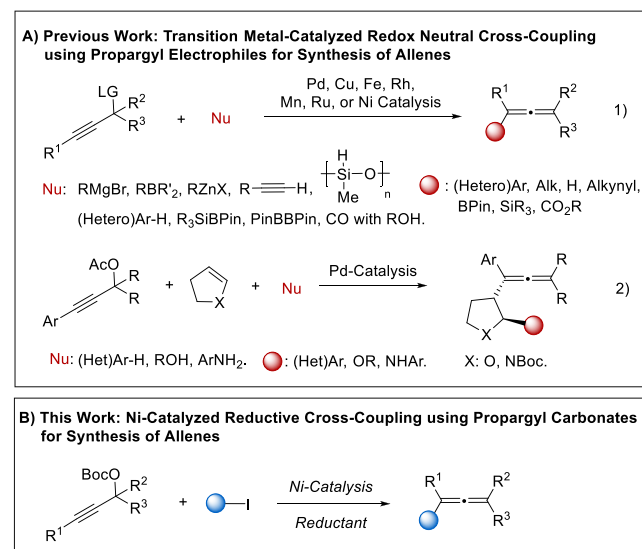
recently, propargyl acetates were utilized as the electrophilic allenyl source in palladium-catalyzed three-component allenylation reactions of dihydrofurans or pyrrolines with (hetero)arenes, alcohols, or anilines as the nucleophiles (Scheme 1A, eq 2).¹³ Furthermore, noncatalytic S_N2' reactions of propargyl electrophiles also provide an alternative approach to preparing various allenes.¹⁴

With the merits of high step-economy and functionality tolerance, nickel-catalyzed cross-electrophile coupling^{15,16} gains increasing attention from organic chemists, and various fundamental organic reactions have been developed relying on this concept, e.g., direct cross-coupling between organo-(pseudo)halides,¹⁷ difunctionalization of alkenes,¹⁸ and ring-opening reactions.¹⁹ However, the use of propargyl electrophiles as allenylating agents in cross-electrophile coupling reactions remains elusive.

Herein, we demonstrate the first application of a cross-electrophile strategy in the synthesis of allenes by reacting propargyl carbonates with alkyl or aryl iodides under the catalysis of nickel with the assistance of a stoichiometric metallic reducing agent. This reaction provides new access to diverse multisubstituted allenes under mild reaction conditions while circumventing the preformation of organometallics.

To realize the target reaction, we initially screened various electrophilic allenylating agents including propargyl carbonates (2a, 2a-3, and 2a-4) and propargyl carboxylates (2a-1 and 2a-2) in the reaction with 1.5 equiv of cyclohexyl iodide (1a)

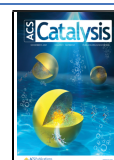
Scheme 1. Transition-Metal-Catalyzed Cross-Coupling Using Propargyl Electrophiles for Synthesis of Allenes



Received: September 9, 2021

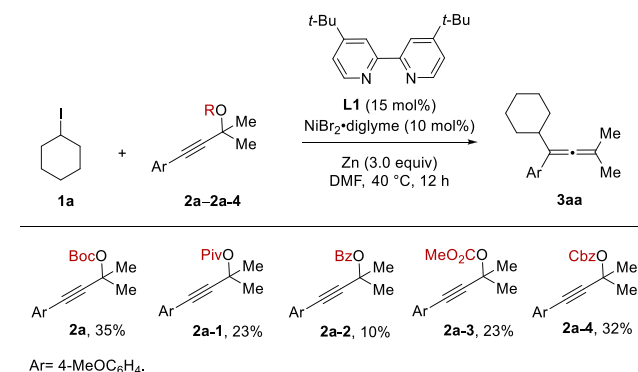
Revised: October 15, 2021

Published: October 19, 2021



under reductive conditions (Table 1). All these reactions were conducted in DMF at 40 °C using NiBr₂·diglyme as a

Table 1. Preliminary Screening of the Electrophilic Allenylating Agents^a



^aReactions were performed on a 0.2 mmol scale of the propargyl electrophiles **2a–2a-4** using 1.5 equiv of cyclohexyl iodide (**1a**), 10 mol % NiBr₂·diglyme, 15 mol % the bipyridine **L1**, and 3.0 equiv of Zn in 0.5 mL of DMF at 40 °C for 12 h. Yields of isolated product through column chromatography.

precatalyst, the bipyridine **L1** as a ligand, and zinc as a reductant. The highest yield (35%) for the desired tetrasubstituted allene **3aa** was achieved in the case of the propargyl *tert*-butyl carbonate **2a** as the precursor.

Inspired by the initial results, we continued to optimize the reaction conditions by varying different reaction parameters (Table 2). Switching the reductant from zinc to manganese gave rise to an improved yield of 57%, wherein the major byproduct turned out to be the trisubstituted allene **3aa'** in 31% yield (entry 1). Of note is that the propargylation of **1a** to yield an alkyne as the product did not occur. No significant decrease in the reaction yield of **3aa** was observed when equimolar cyclohexyl iodide (**1a**) was used (entry 2). Notably, the amount of the byproduct **3aa'** could be effectively reduced in this case. Similar yields could be achieved in other polar solvents like DMA and NMP (entries 3 and 4). In contrast, no desired reaction occurred when performing the reaction in THF (entry 5). Next, a brief ligand screening was undertaken. In the case of 2,2'-bipyridine (**L2**) and the phenanthrolines **L3** and **L4**, the product **3aa** was afforded in slightly diminished efficiency (entries 6–8). When the terpyridine **L5** was employed as the ligand, the formation of the allene **3aa** was almost shut down (entry 9). By the use of 1.0 equiv of K₂HPO₄ as an additive, an elevated yield for **3aa** could be achieved (entry 10). Finally, other nickel(II)-precatalysts were investigated (entries 11–13), and the best result (71% yield) was delivered in the case of NiBr₂(dme) (entry 13).

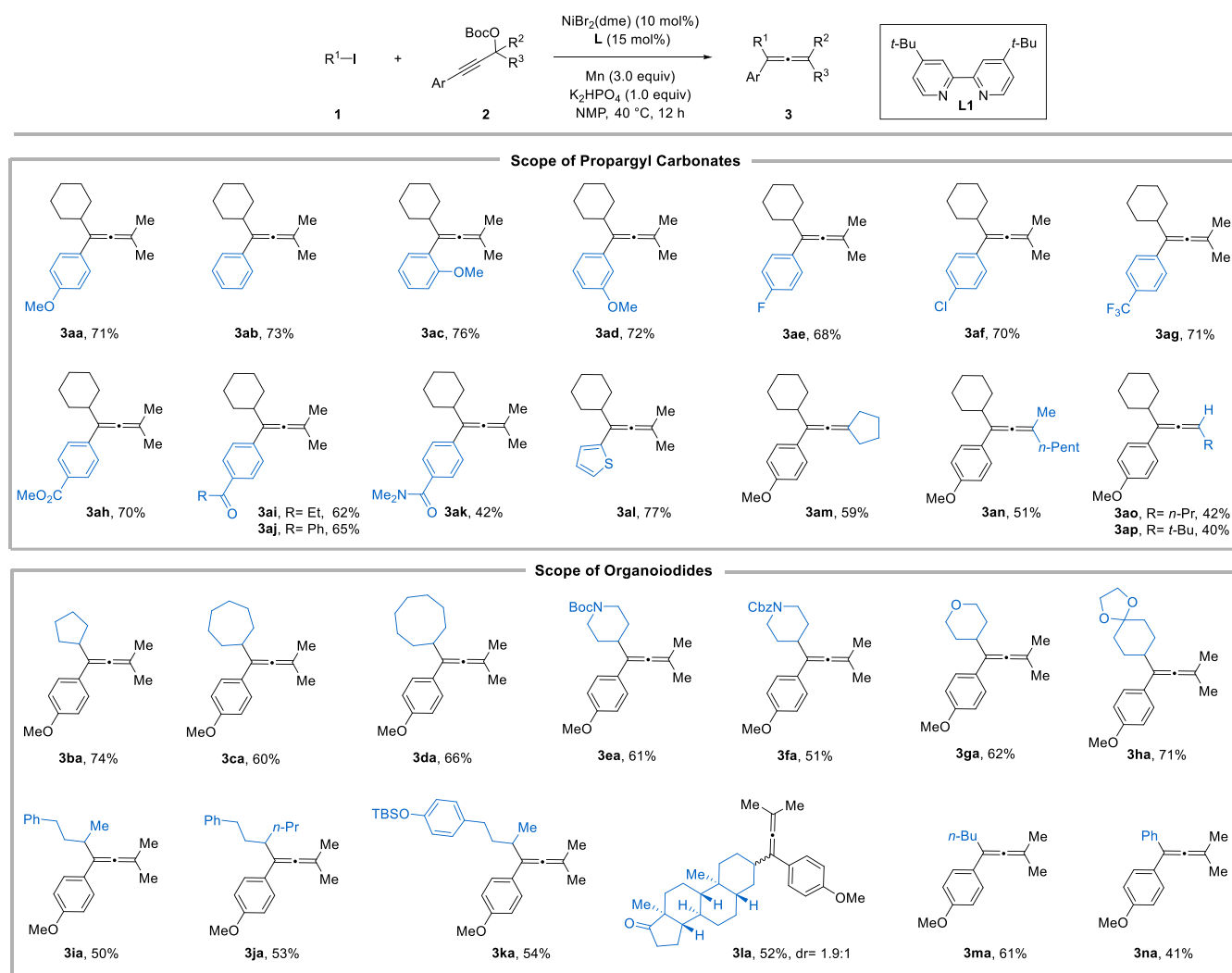
With the optimal reaction conditions in hand, we set out to explore the substrate scope of this reductive reaction for the synthesis of allenes (Table 3). First, various substituents on the phenyl ring of propargyl carbonates (**2a–k**) were examined in the reactions with cyclohexyl iodide (**1a**). Gratifyingly, both electron-donating and electron-withdrawing groups were well-tolerated, providing the products **3aa–ak** in moderate to good yields. Notably, organometallic-sensitive ester (**3ah**) and ketone (**3ai** and **3aj**) proved to be compatible in our reaction. Furthermore, the thienyl-substituted propargyl carbonate **2l** turned out to be a suitable precursor, furnishing the product

Table 2. Optimization of the Reaction Conditions^a

entry	ligand	Ni-precatalyst	solvent	yield (%) ^b
1 ^c	L1	NiBr ₂ ·diglyme	DMF	57 (31)
2	L1	NiBr ₂ ·diglyme	DMF	54 (15)
3	L1	NiBr ₂ ·diglyme	DMA	47 (4)
4	L1	NiBr ₂ ·diglyme	NMP	55 (4)
5	L1	NiBr ₂ ·diglyme	THF	0 (0)
6	L2	NiBr ₂ ·diglyme	NMP	40 (19)
7	L3	NiBr ₂ ·diglyme	NMP	38 (16)
8	L4	NiBr ₂ ·diglyme	NMP	47 (15)
9	L5	NiBr ₂ ·diglyme	NMP	trace (0)
10 ^d	L1	NiBr ₂ ·diglyme	NMP	66 (2)
11 ^d	L1	Ni(BF ₄) ₂ ·6H ₂ O	NMP	0 (0)
12 ^d	L1	NiI ₂	NMP	58 (<1)
13 ^d	L1	NiBr ₂ (dme)	NMP	71 (<1)

^aReactions were performed on a 0.2 mmol scale of the propargyl carbonate **2a** using 1.0 equiv of cyclohexyl iodide (**1a**), 10 mol % Ni-precatalyst, 15 mol % ligand **L**, and 3.0 equiv of Mn in 0.5 mL of solvent at 40 °C for 12 h. ^bYields of the isolated product **3aa** through column chromatography. NMR yields of the byproduct **3aa'** are given in parentheses. ^cReaction was performed with 2.0 equiv of cyclohexyl iodide (**1a**). ^dReaction was performed with 1.0 equiv of K₂HPO₄.

3al in 77% yield. The propargyl carbonate **2m** with a cyclopentane scaffold was also well-accommodated, and the corresponding product **3am** was obtained in 59% yield. Moreover, the unsymmetric substituted tertiary propargyl carbonate **2n** was successfully used as the coupling partner, furnishing the allene **3an** in 51% yield. Secondary propargyl alcohol derivatives **2o** and **2p** were also competitive substrates for this nickel-catalyzed reaction. In these cases, the products **3ao** and **3ap** were delivered in relatively low yields, because the corresponding, less bulky secondary propargyl carbonates are more prone to undergo the homocoupling reaction. Remarkably, the formation of the cross-coupling products at the propargylic position was not observed; these were found to be the major products in the nickel-catalyzed redox-neutral coupling between propargyl electrophiles and organozincs.²⁰ The positive result of the reaction using the *t*-Bu-substituted propargyl carbonate **2p** lacking β-H excludes a reaction pathway engaging the in-situ-generated enyne for this nickel-catalyzed allene synthesis. One limitation of our method was noticed in the case of aliphatic propargyl carbonates, which did not lead to the formation of allene products. At this juncture, we turned out attention to study the scope of alkyl iodides. To our delight, the reactions using cyclic or acyclic secondary alkyl iodides (**1b–k**) proceeded smoothly under the standard reaction conditions, affording the products **3ba–ka** in moderate to good efficiency. Remarkably, the structurally complex alkyl iodide **1l** derived from a steroid posed no problem, giving two diastereomers (**3la**) in 53% combined yield, which could be simply separated through column

Table 3. Evaluation of the Substrate Scope of Reductive Cross-Coupling between Organoiodides and Propargyl Carbonates^a

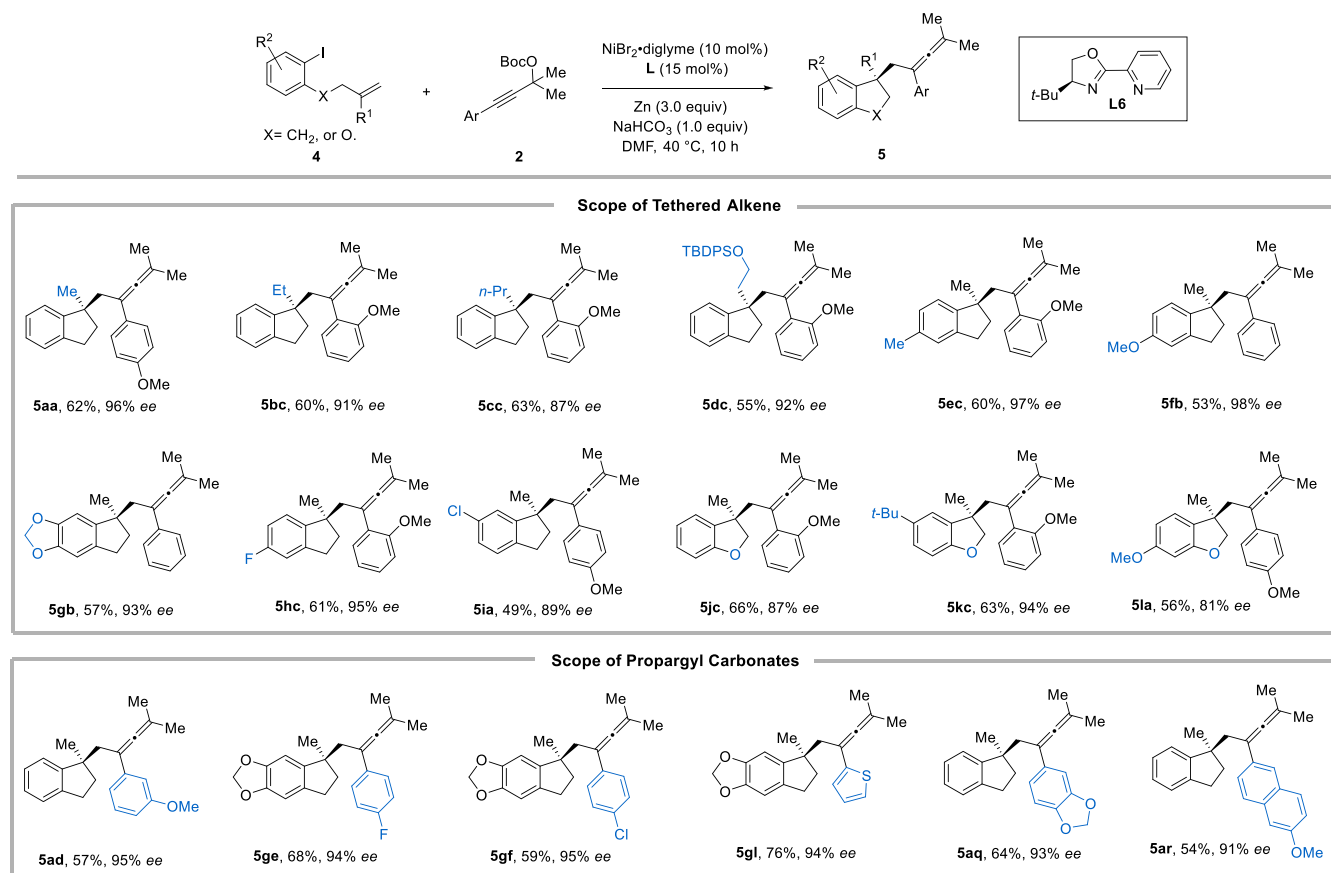
^aReactions were performed on a 0.2 mmol scale of the propargyl carbonates **2** using 1.0 equiv of organoiodides **1**, 10 mol % NiBr₂·(dme), 15 mol % the bipyridine **L1**, 3.0 equiv of Mn, and 1.0 equiv of K₂HPO₄ in 0.5 mL of NMP at 40 °C for 12 h. Yields of the isolated products after column chromatography.

chromatography. Primary alkyl iodides exemplified by *n*-butyl iodide (**1m**) also proved to be pertinent precursors, affording the product **3ma** in 61% yield. Unfortunately, the reactions employing tertiary alkyl halides failed to generate the desired allene products. In addition, our method is applicable for the use of phenyl iodide (**1n**) as the coupling partner, wherein compound **3na** was formed in 41% yield.

In recent years, a breakthrough has been achieved in nickel-catalyzed asymmetric reductive dicarbofunctionalizations of tethered alkenes with two distinct electrophiles, allowing for construction of a chiral benzene-annulated framework in a highly enantioselective manner.^{18a–i} Following this strategy, the successfully installed functional groups across the pendant olefinic unit include aryl,^{18a} alkyl,^{18b} alkenyl,^{18c} benzyl,^{18d} allyl,^{18e,f} acyl,^{18g} carbamoyl,^{18g} and carboxyl.^{18h,i} Having realized the nickel-catalyzed direct cross-coupling between organoiodides and propargyl carbonates shown above, we attempted to develop an unprecedented asymmetric olefin aryl-allenylation using propargyl carbonates as the electrophilic allenylating source. To our delight, high asymmetric induction could be achieved in the target reaction when using the chiral

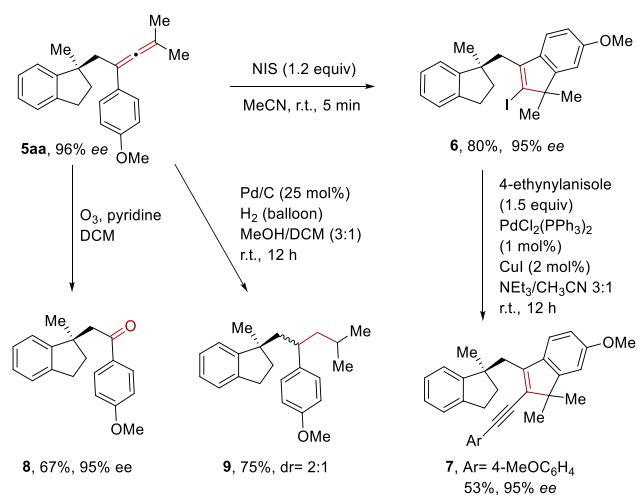
pyrox **L6** as the ligand under the modified reaction conditions: zinc as the reductant, NaHCO₃ as the additive, and DMF as the solvent. The substrate scope of this aryl-allenylation reaction was also investigated (Table 4). In general, diverse chiral indanes and dihydrobenzofurans (**5**) bearing an allenyl-substituted quaternary stereocenter were synthesized in moderate to good efficiency and with excellent enantioselectivities. The bulkiness of the geminal substituent of the incorporated alkene (**4a–c**) did not influence the reaction outcome profoundly. Various substituents on the phenyl ring of tethered aryl iodides (**4d–l**) were tolerable. Furthermore, aryl and heteroaryl-substituted propargyl carbonates (**2d–f**, **2l**, **2q**, and **2r**) were competent coupling partners.

Next, various product derivatizations using the allene moiety as the functional handle were carried out (Scheme 2). First, N-iodosuccinimide-mediated aryl-iodination of the tethered allene of compound **5aa** enabled the construction of the 2-iodoindene framework (**6**) in 80% yield,²¹ which was subsequently subjected to palladium-catalyzed Sonogashira coupling, delivering the product **7** in 53% yield. Moreover, the conversion of allene to ketone was achieved via ozonolysis,

Table 4. Evaluation of the Substrate Scope of Asymmetric Aryl-Allenylation of Tethered Alkenes^a

^aReactions were performed on a 0.2 mmol scale of the propargyl carbonates **2** using 2.0 equiv of aryl-iodide-tethered alkenes **4**, 10 mol % NiBr₂·(dme), 15 mol % the pyrox **L6**, 3.0 equiv of Zn, and 1.0 equiv of NaHCO₃ in 0.5 mL of DMF at 40 °C for 10 h. Yields of the isolated products after column chromatography. Enantiomeric excesses were determined by HPLC analysis on a chiral stationary phase.

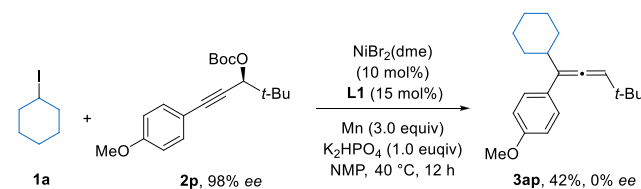
Scheme 2. Derivatizations of the Aryl-Allenylation Product



furnishing the ketone **8** in 67% yield. In addition, Pd/C-catalyzed hydrogenation of the allene **5aa** allowed the formation of compound **9** in 75% yield with a diastereomeric ratio of 2:1.

To gain insight into the stereochemical course of this nickel-catalyzed reaction, an enantioenriched (98% ee) propargyl carbonate (**2p**)²² was employed as the precursor (Scheme 3).

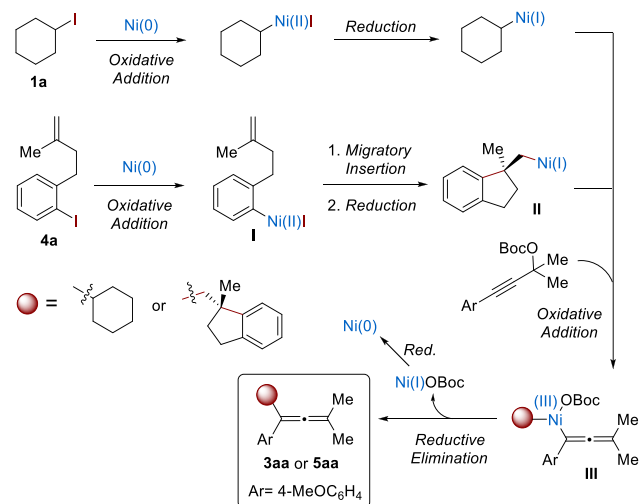
Scheme 3. Study of the Stereochemical Course



In this case, no chirality transfer occurred, since the allene product **3ap** was formed in racemic form, indicating that racemization occurred in the oxidative addition of the propargyl carbonate to low-valent nickel.^{20b,c}

On the basis of the previous reports, we proposed a plausible reaction mechanism involving sequential oxidative additions for this nickel-catalyzed reductive cross-coupling toward the synthesis of allenes exemplified by the reaction to form **3aa** or **5aa** (Scheme 4). Initially, a Ni(0) species is generated under the reductive condition, which undergoes oxidative addition with the organoiodides **1a** or **4a**, to afford the direct oxidative adduct CyNi(II)I or the aryl Ni(II) complex I. In the former case, CyNi(II)I is instantly reduced to CyNi(I). In the latter case, facially selective migratory insertion precedes the reduction.^{18b,23} Subsequently, the propargyl carbonate **1a** conducts the second oxidative addition to CyNi(I) or the σ -alkyl-Ni(I) intermediate II. Facile reductive elimination from the resultant allenyl Ni(III) complexes III affords the allene

Scheme 4. Proposed Reaction Mechanism



product **3aa** or **5aa**. Finally, the Ni(0) species is regenerated for the next catalytic cycle via zinc- or manganese-mediated reduction of Ni(I)OBoc.

In conclusion, we developed a nickel-catalyzed cross-electrophile coupling between organoiodides and propargyl carbonates, enabling the synthesis of diverse multisubstituted allenes without tedious preformation of organometallics. Furthermore, we successfully utilized propargyl carbonates as the allenyl source in the nickel-catalyzed asymmetric reductive aryl-allenylation of tethered alkenes, to provide a method to approach chiral indanes and dihydrobenzofurans containing an allenyl-substituted quaternary stereogenic center in a highly enantioselective fashion.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acscatal.1c04143>.

Representative experimental procedures and necessary characterization data for all new compounds (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work is supported by “1000-Youth Talents Plan” starting up funding, the National Natural Science Foundation of China (21772183, 22071230), the Fundamental Research Funds for the Central Universities (WK2060190086), as well as the University of Science and Technology of China.

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